EVERY FINITE SEPARABLE EXTENSION OF THE COMPLETION OF A DISCRETE VALUATION FIELD IS OBTAINED BY BASE CHANGE FROM A FINITE SEPARABLE EXTENSION OF THE FIELD

THE STACKS PROJECT AUTHORS

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Editorial note. The author states at proof lines 33825-33827 that the proof works more generally for Noetherian local rings whose completion is a local domain of dimension one; that remark is part of the counted proof and is delivered as written.

Editorial note. The parenthetical remark on the theorem of the primitive element at proof line 33831 is the author's own and is retained inside the counted span.

Editorial note. The proof refers to three earlier lemmas of this chapter, two about completions of discrete valuation rings and one being Krasner's lemma; all three are carried as navigation stubs with their roles recorded in the reasons.

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Allowed standard prerequisites (editorial).

- The completion of a discrete valuation ring is again a discrete valuation ring, in particular a domain. used at proof lines 33823-33825. Makes the completed ring a complete local domain of dimension one, which is the setting the argument is carried out in.
- Krasner's lemma: a polynomial with a simple root survives a sufficiently small perturbation of its coefficients, the new root being close to the old one in the maximal-adic topology. invoked at proof lines 33841-33844. Produces a root beta in B of the approximating polynomial P_1 and allows beta to be chosen so that theta - beta lies in pi B.

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- Nakayama's lemma applied to the finite module B over a complete local domain, and the equality of degrees making a surjective map between the two quotients an isomorphism. used at proof lines 33845-33849. Shows that the comparison map from A-completion[t] over P_1 to B is surjective and hence an isomorphism, which identifies the base change of L with M.

Source reference. Original lemma 07NY: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10668–10668. The reference is one of the two applied at proof lines 33823-33824 to know that the completion of A is again a discrete valuation ring; the construction of the extension is inside the counted span.

Source reference. Original lemma 07NV: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10624–10624. The reference is the second of the two applied at proof lines 33823-33824 for the discreteness of the completion; everything done with that fact is inside the counted span.

Source reference. Original lemma 09EI: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33748–33748. The reference is applied at proof lines 33841-33844 to obtain the root beta of the approximating polynomial P_1 in B with theta - beta in pi B; the construction of P_1 and the Nakayama comparison that consume it are inside the counted span.

Lemma 0.2. *Let A be a discrete valuation ring with field of fractions K. Let $A^\wedge$ be the completion of A with fraction field $K^\wedge$. If $M/K^\wedge$ is a finite separable extension, then there exists a finite separable extension L/K such that $M = K^\wedge \otimes_K L$.*

Proof. Note that $A^\wedge$ is a discrete valuation ring too (by Lemmas 07NY and 07NV). In particular $A^\wedge$ is a domain. The proof will work more generally for Noetherian local rings A such that $A^\wedge$ is a local domain of dimension 1.

Let $\theta \in M$ be an element that generates M over $K^\wedge$. (Theorem of the primitive element.) Let $P(t) \in K^\wedge[t]$ be the minimal polynomial of θ over $K^\wedge$. Let $\pi \in \mathfrak{m}_A$ be a nonzero element. After replacing θ by $\pi^n\theta$ we may assume that the coefficients of $P(t)$ are in $A^\wedge$. Let $B = A^\wedge[\theta] = A^\wedge[t]/(P(t))$. Note that B is a complete local domain of dimension 1 because it is finite over A and contained in M. Since M is separable over K the element θ is not a root of the derivative of P. For any integer n we can find a monic polynomial $P_1 \in A[t]$ such that $P - P_1$ has coefficients in $\pi^n A^\wedge[t]$. By Krasner's lemma (Lemma 09EI) we see that P_1 has a root β in B for n sufficiently large. Moreover, we may assume (if n is chosen large enough) that $\theta - \beta \in \pi B$. Consider the map $\Phi : A^\wedge[t]/(P_1) \to B$ of $A^\wedge$-algebras which maps t to β. Since $B = \pi B + \sum_{i<\deg(P)} A^\wedge\theta^i$, the map Φ is surjective by Nakayama's lemma. As $\deg(P_1) = \deg(P)$ it follows that Φ is an isomorphism. We conclude that the ring extension $L = K[t]/(P_1(t))$ satisfies $K^\wedge \otimes_K L \cong M$. This implies that L is a field and the proof is complete. □